

Labor Input Responses to Ownership Changes: Evidence from U.S. Nursing Homes

Joseph White

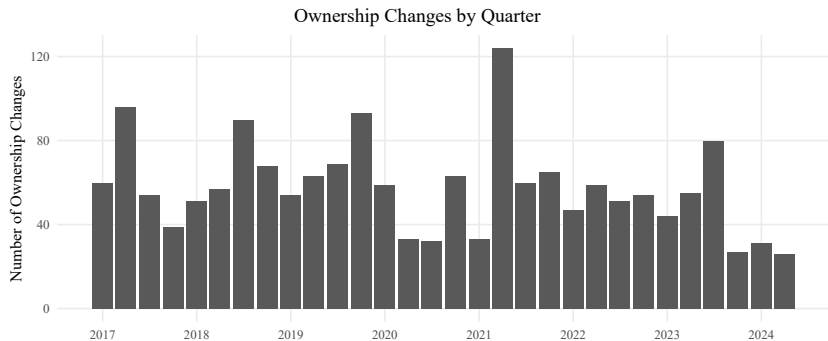
Miami University

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Motivation: Ownership Changes Beyond Market Structure

- Ownership changes are very common
 - ▶ (1) Mergers and Acquisitions
 - ▶ (2) New Management
- This second type of changes does **not** change the set of providers in a market.
- Instead, effects operate through **internal production decisions**

Ownership Changes Over Time (Quarterly)



Why Nursing Homes?

- There are about 15,000 nursing homes in the U.S. that serve over 4 million people annually
- 56% of adults aged 57 to 61 will spend at least one night in a nursing home during their lifetime
- Nursing home care is labor-intensive and heavily regulated.
- **Nursing labor is the main input** into care and a major operating cost.
- Staffing is a natural adjustment margin after ownership change, but it's constrained by:
 - ▶ compliance,
 - ▶ resident outcomes,
 - ▶ demand/occupancy

Staffing is Important

- Staffing is the primary input into nursing home care, so ownership changes can affect quality even when market structure is unchanged.
- Higher staffing levels are linked to better quality measures in nursing homes, including fewer deficiency citations ([Harrington et al., 2000](#)).
- Causal evidence shows that increasing nurse staffing improves measured nursing home quality ([Lin, 2014](#)).
- If ownership changes shift staffing decisions, they can translate into meaningful differences in care quality.

- **Research question:** How do nurse staffing levels respond to ownership changes in nursing homes?
 - ▶ Do responses differ across staff type
 - ▶ Do responses differ if a nursing home is part of a chain or not?
 - ▶ Do responses differ pre vs post-pandemic?

Nursing Staff Types and Staffing Measurement

- **Nursing homes staff care using three main nursing roles:**
 - ▶ **RN (Registered Nurses):** clinical assessment, care planning, meds, supervision
 - ▶ **LPN (Licensed Practical Nurses):** routine clinical tasks, meds under RN oversight
 - ▶ **CNA (Certified Nursing Assistants):** direct daily care
 - ▶ **Total nursing:** RN + LPN + CNA
- **Staffing measure: Hours Per Patient Day (HPPD)**
 - ▶ Standardizes staffing by resident census (staffing intensity, not raw hours)
 - ▶ Lets us compare facilities and months on a common scale
- Ownership change can affect both **levels** (total HPPD) and **mix** (RN/LPN/CNA composition) which sets up cost-quality tradeoff.

The Cost–Quality Tradeoff in Staffing

- Owners can raise quality through two channels:
 - ▶ **Quantity (levels):** increasing total nursing hours (HPPD)
 - ▶ **Skill mix:** shifting hours toward more clinically trained staff (e.g., more RN oversight)
- **Both margins raise costs** because nursing labor is a major operating expense.
- **But short-run revenues are relatively fixed** (public payers, limited pricing flexibility), so staffing choices map into a **cost–quality tradeoff**.
- **Implication:** an ownership change that shifts objectives or constraints can move a facility along this tradeoff. ([Park and Werner, 2011](#); [Bowblis and Brunt, 2025](#))

Why Staffing Could Rise or Fall After a Transition

- Ownership changes can shift **constraints** and **objectives** $\Rightarrow$ staffing effects are ambiguous.
- **Scenario 1: Cost pressure $\Rightarrow$ staffing falls**
 - ▶ Debt service / tighter targets $\Rightarrow$ cut labor costs (lower HPPD or cheaper mix)
- **Scenario 2: Quality/compliance push $\Rightarrow$ staffing rises**
 - ▶ Invest in staffing to improve oversight / survey performance (higher HPPD or better mix)
- **Scenario 3: Reorganization $\Rightarrow$ mix changes**
 - ▶ Standardized staffing model shifts RN/LPN/CNA composition; total hours may be similar
- **Bottom line:** competing predictions $\Rightarrow$ **empirical question.** (Gupta et al., 2021; Holmes, 1996)

Data Sources and Unit of Observation

- **Healthcare Cost Report Information System (HCRIS)**
 - ▶ Annual cost report information
 - ▶ Contains information about costs, revenue, financial data and dates of ownership change
- **Nursing Home Compare (NHC) Archive**
 - ▶ Contains ownership information as well as facility characteristics
- **Payroll Based Journal (PBJ)**
 - ▶ Payroll-reported nurse staffing hours by staff type
 - ▶ Used to construct outcome variables
- **Facility-month panel, 48 contiguous states, Jan 2017 – Jun 2024**

Identifying Ownership Changes: Step 1

- **Starting point (HCRIS):** includes a variable indicating **whether and when** an ownership change occurred.
- **Challenge:** ownership is often **layered** (organizations owning organizations), so an administrative flag may not map cleanly to an **economic change in control**.
- HCRIS might not contain all ownership changes, or it may contain those with no meaningful economic change, so we have to verify using the NHC Archive data.

Ownership Level Challenge

- We identify ownership changes at the **highest level of ownership present**

Ownership Level	Type	Owner Name	Share
<i>Direct ownership</i>			
Organization	LLC	BREDLEGS HOLDINGS LLC	25%
Organization	LLC	ROCK HILL ACQUISITION III LLC	75%
<i>Indirect ownership</i>			
Individual	Person	LALLY, THOMAS	25%
Individual	Person	LITT, ALAN	25%
Individual	Person	LITT, JONATHAN	25%
Individual	Person	STURM, JOSHUA	25%

Identifying Ownership Changes: Step 2

- **Validate & date using NHC:**

- ▶ Track ownership shares over time within the highest reported ownership level
- ▶ Define a change when **majority control (50%)** transfers between $t - 1$ and t
- ▶ We do not flag changes driven by name changes or within-family reclassification

- **Example from the data:**

- ▶ *Before:*

- ★ ESTES JAMES (84%)
- ★ BURCHFIELD JOHN (10%)
- ★ LEE CLAUDE (6%)

- ▶ *After:*

- ★ JAMES N ESTES JR FAMILY DYNASTY TR NO 2 (50%)
- ★ JENNIFER E AGEE FAMILY DYNASTY TR NO 2 (50%)

- ▶ We do **not** treat this as a meaningful ownership change

Ownership Change Agreement: NHC vs HCRIS

- **Step 3: Validate the ownership-change flag.**
- We keep facilities with:
 - ▶ **(0,0)**: no changes in either source, or
 - ▶ **(1,1)**: exactly one change in each source with dates within **6 months**.
- **Event month**: the first month NHC indicates a majority-control transfer.

Cross-tab of Ownership Changes

Count of changes in NHC	2+	1,137 (8.5%)	1,377 (10.3%)	535 (4.0%)
	1	1,874 (14.0%)	1,589 (11.9%)	101 (0.8%)
	0	6,217	499	15

Dependent Variables: Nurse Staffing (PBJ) and Analysis Sample

- We study four staffing measures:
 - ▶ RN, LPN, CNA, and total HPPD
- Outcome is **hours per patient day (HPPD)**: staffing intensity standardized by resident census

$$\text{HPPD}_{s,i,t} = \frac{H_{s,i,t}}{\text{PatientDays}_{i,t}}$$

- **CMS implausible staffing exclusions:**
 - ▶ RN HPPD = 0 **and** LPN HPPD = 0
 - ▶ Total HPPD < 1.5 or > 12
 - ▶ CNA HPPD > 5.25
- **Additional restrictions:** beds < 15; hospital-based facilities
- **Final sample:** 7,806 facilities; 632,231 facility-months; 1,589 treated ($\approx 20\%$)

Empirical Strategy: Event-Study Difference-in-Differences

- **Design:** staggered-adoption DiD around ownership transitions
[Goodman-Bacon, 2021](#)
- **Variation:** within-facility staffing changes relative to facilities *not yet treated*
- **Event study regression:**

$$Y_{it} = \sum_{\tau=-24}^{24} \beta_{\tau} \mathbf{1}\{event_time_{it} = \tau\} \cdot Treat_i + X'_{it}\gamma + \alpha_i + \lambda_t + \varepsilon_{it}.$$

- **Key objects:**

- ▶ Y_{it} : staffing outcome in **HPPD** (RN, LPN, CNA, Total)
- ▶ β_{τ} : effect at event time τ (relative to the reference period)
- ▶ α_i facility FE; λ_t month FE; X_{it} controls (beds, occupancy, payer mix, ownership type, chain, acuity)
- ▶ SEs clustered at the facility level

Empirical Strategy: TWFE Difference-in-Differences

- **Goal:** summarize the **average** staffing change after an ownership transition
- **TWFE DiD regression:**

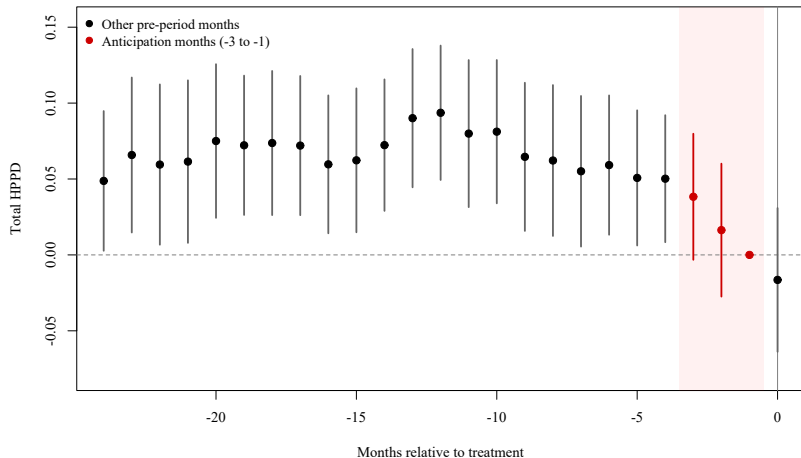
$$Y_{it} = \beta Post_{it} + X'_{it}\gamma + \alpha_i + \lambda_t + \varepsilon_{it}$$

- **Key objects:**
 - ▶ $Post_{it} = 1$ in all months after a change (= 0 before)
 - ▶ β : **average post-change effect** on staffing (HPPD)
 - ▶ Same fixed effects and controls as the event-study specification
 - ▶ SEs clustered at the facility level

Identification: Parallel Trends

- **Parallel trends:** absent an ownership change, treated facilities would have followed the same staffing trends as facilities not yet treated.
- **Why anticipation is plausible in ownership transitions**
 - ▶ Changes are not instantaneous: negotiations and operational handoffs precede the recorded month
 - ▶ Staffing can adjust during planning (turnover, hiring freezes, renegotiated contracts)
 - ▶ Recorded dates may reflect administrative/legal timing, not first operational change
- **Implication:** months just before $\tau = 0$ may violate parallel trends.

Evidence of violation in the Pre-Period



Main Specification: Donut Event Study + Pre-trend Test

- **Donut event study:** mitigate this influence by excluding months just before the recorded transition.
- **Implementation**
 - ▶ Exclude $\tau \in \{-3, -2, -1\}$
 - ▶ Use $\tau = -4$ as the reference period
- **Parallel Trends:** joint Wald test of pre-period leads

$$H_0 : \beta_\tau = 0 \quad \forall \tau < 0$$

- **Takeaway:** full pre-window shows evidence of pre-trends; the donut specification is more consistent with parallel trends.

Parallel Trends Tests (Joint Wald) — Total Staffing

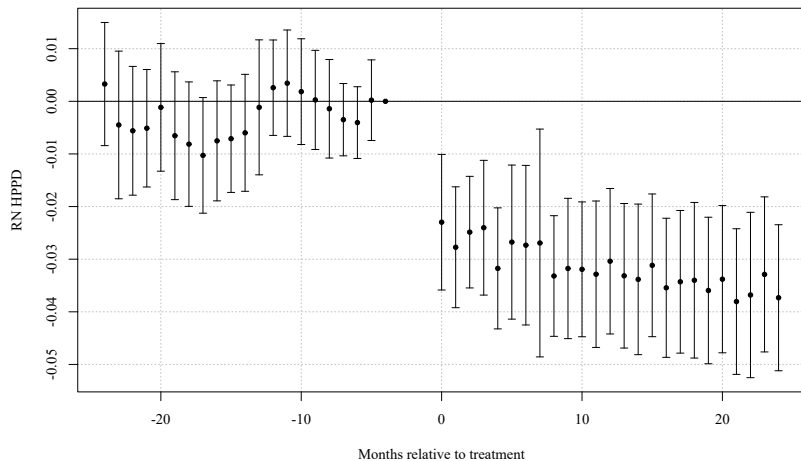
	Total (Levels)	Total (Logs)
2 Year Full Pre-Window	0.0006	0.0012
2 Year Window with Donut	0.2046	0.3268
1 Year Window with Donut	0.6387	0.6102

Notes: Joint Wald χ^2 test of $H_0 : \beta_\tau = 0$ for all pre-treatment coefficients. Cells report p-value.

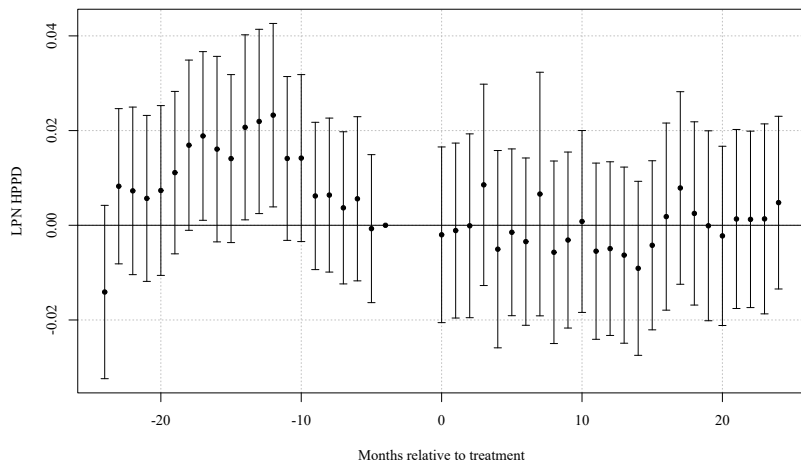
Donut excludes $\tau \in \{-3, -2, -1\}$; reference period is $\tau = -4$.

Full outcome-by-staff-type table available in appendix.

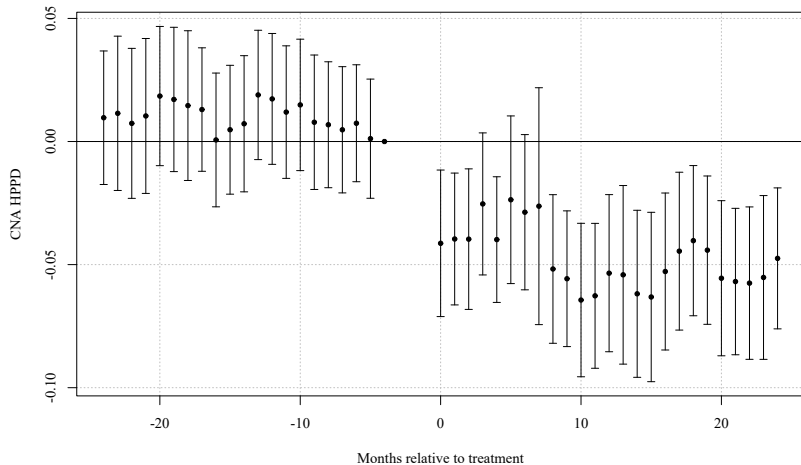
Main Results: Event Study (RN HPPD)



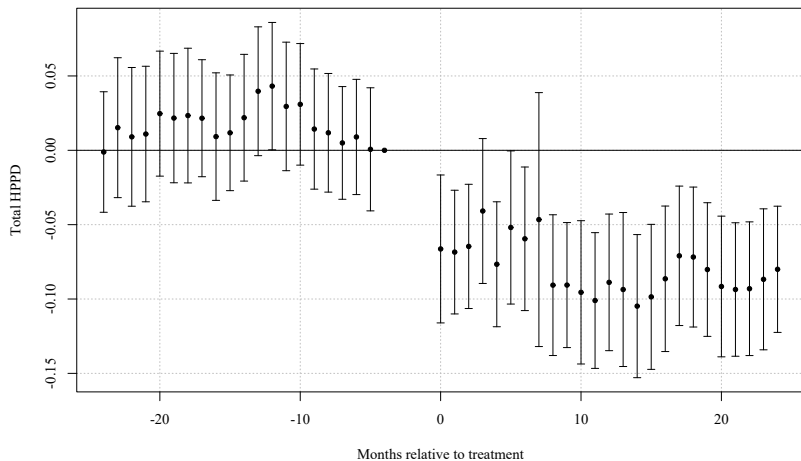
Main Results: Event Study (LPN HPPD)



Main Results: Event Study (CNA HPPD)



Main Results: Event Study (Total Nursing HPPD)



TWFE DiD — Baseline (RN)

	RN	LPN	CNA	Total
HPPD	−0.032*** (0.005)	0.000 (0.006)	−0.054*** (0.009)	−0.086*** (0.013)
Log(HPPD)	−0.092*** (0.016)	0.003 (0.009)	−0.028*** (0.005)	−0.026*** (0.004)

Notes: Each cell reports the coefficient on *post* with two-way clustered SEs (facility and month) in parentheses. Sample: without anticipation ($N = 628,107$).

TWFE DiD — Baseline (LPN)

	RN	LPN	CNA	Total
HPPD	−0.032*** (0.005)	0.000 (0.006)	−0.054*** (0.009)	−0.086*** (0.013)
Log(HPPD)	−0.092*** (0.016)	0.003 (0.009)	−0.028*** (0.005)	−0.026*** (0.004)

Notes: Each cell reports the coefficient on *post* with two-way clustered SEs (facility and month) in parentheses. Sample: without anticipation ($N = 628,107$).

TWFE DiD — Baseline (CNA)

	RN	LPN	CNA	Total
HPPD	-0.032*** (0.005)	0.000 (0.006)	-0.054*** (0.009)	-0.086*** (0.013)
Log(HPPD)	-0.092*** (0.016)	0.003 (0.009)	-0.028*** (0.005)	-0.026*** (0.004)

Notes: Each cell reports the coefficient on *post* with two-way clustered SEs (facility and month) in parentheses. Sample: without anticipation ($N = 628,107$).

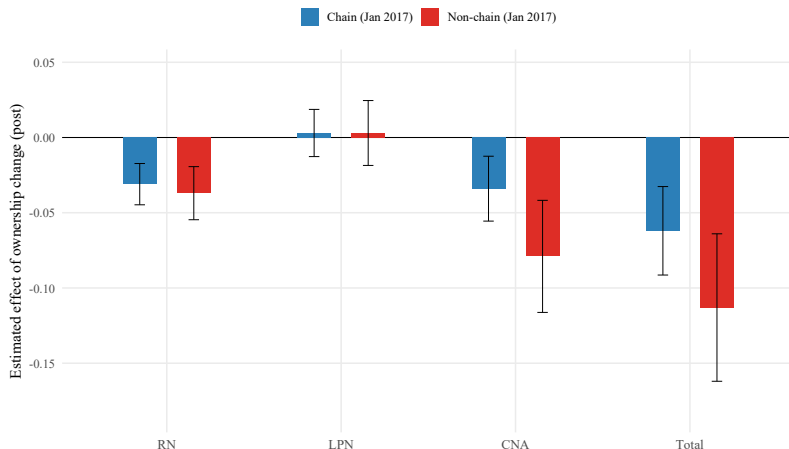
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Notes: Each cell reports the coefficient on *post* with two-way clustered SEs (facility and month) in parentheses. Sample: without anticipation ($N = 628,107$).

Heterogeneity: Chain vs Non-Chain Facilities

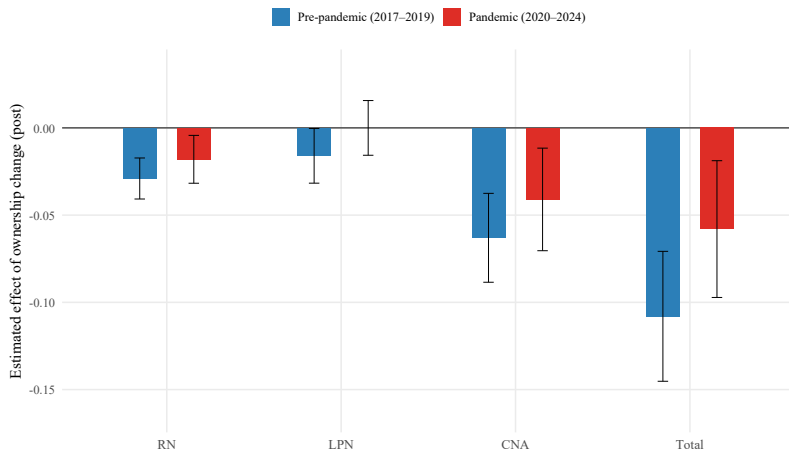
TWFE Post Estimates: Chain vs Non-chain (HPPD, Without Anticipation)



Staffing declines are larger for non-chain facilities, nearly double for CNA and Total.

Heterogeneity: Pre vs Post-Pandemic

TWFE Post Estimates: Pre-pandemic vs Pandemic (HPPD, Without Anticipation)



Large pre-pandemic declines, and post-pandemic effects are almost cut in half.

- **Control-group**

- ▶ Re-estimate using **not-yet-treated** controls only.
- ▶ Re-estimate using **never-treated** facilities only.
- ▶ **Stacked DiD** (cohort-by-cohort).

- **Treatment timing / definition.**

- ▶ Shift event date by ± 1 month.
- ▶ NHC vs HCRIS

• Donut Hole Specification

- ▶ Baseline uses a **donut**: drop $\tau \in \{-3, -2, -1\}$ (reference $\tau = -4$).
- ▶ **Wider donut**: drop $\tau \in \{-4, -3, -2, -1\}$.
- ▶ **Smaller donut**: drop only $\tau \in \{-2, -1\}$.
- ▶ Estimates are unchanged in sign, magnitude, and significance.

• Reporting gaps in staffing data.

- ▶ Restrict sample to facilities with smaller gaps between observations (exclude gaps > 6 , > 3 , > 1 , and > 0 months).
- ▶ Estimates are nearly identical to baseline.

References I

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